

QuantumSepsis Shield

*An Adversarially-Safe Multi-Agent Hybrid Quantum–Classical System for
Early Sepsis Trajectory Detection in Resource-Constrained ICU Environments*

Theme: Innovating the Quantum Frontier

1. Introduction

Sepsis is the life-threatening organ dysfunction caused by an uncontrolled host response to infection, and it is one of the leading causes of preventable in-hospital deaths worldwide, resulting in the deaths of 11 million people every year, with almost 7.8 million cases reported in India every year. Despite decades of research, sepsis is still an underdiagnosed condition in its early stages, when it is most treatable.

The problem is that sepsis does not manifest as an acute, singular event, but rather as an insidious, multivariate physiological progression of over 12 simultaneous variables, where the heart rate increases gradually, the temperature increases, the blood pressure drops gradually, but none of the individual monitors in the intensive care unit (ICU) go off, and by the time the picture is obvious, it is already too late, resulting in an increased risk of death by 7% for every hour of delay.

Moreover, the current diagnostic approach is fundamentally sequential, where the physician orders the complete blood count, waits for the results, then orders the lactate, then the procalcitonin, and so forth, taking 4 to 6 hours before the final treatment bundle is initiated. While existing AI and ML tools have pushed sepsis prediction forward, they are largely built as retrospective, binary classifiers. This makes them poorly suited for the realities of Indian ICUs, which desperately need real-time intervention despite having scarce training data.

We've certainly seen classical models perform well when tested on the massive MIMIC-III database. To clarify, specific implementations such as the autotriage algorithm (Nemati) and LSTM networks (Sherpf) have reliably distributed AUROCs between 0.81 and 0.85. A comprehensive review by Moore et al. This trend was confirmed in 2021, with 22 separate evaluations revealing predictive scores ranging from 0.74 to 0.92. These models consistently perform worse when trained on fewer than 1000 samples. In a typical Indian intensive care unit, which may record only 200 to 500 sepsis cases annually, hunger for data is a fatal flaw.

QuantumSepsis Shield offers a radical shift to bypass this limitation. By identifying deterioration in the patient's condition within the first hour, the system triggers all downstream clinical actions simultaneously. By running these tests in parallel, the system eliminates the standard 6-hour diagnostic delay and replaces it with real-time, proactive monitoring.

2. Literature Survey

The clinical basis for this work is rooted in the Sepsis-3 consensus definitions (Singer et al., JAMA, 2016), which identify organ dysfunction as the key feature of sepsis and propose the Sequential Organ Failure Assessment (SOFA) score as the new standard for diagnosis. Sepsis-3 also highlights the shortcomings of the previous SIRS-based definitions, which led to the development of data-driven solutions that can detect organ dysfunction before the onset of overt clinical symptoms.

The global burden of sepsis was quantitatively assessed in a landmark study by Rudd et al. (The Lancet, 2020), which found that there were 48.9 million new cases and 11 million sepsis-related deaths in 2017, corresponding to 19.7% of total global mortality. This study also highlighted the large burden in low- and middle-income countries, including India, which would benefit from the use of automated solutions due to a lack of intensive care unit (ICU) infrastructure and diagnostic turnaround times compared to high-income countries.

We validate our temporal models using the MIMIC-III database (Johnson et al., 2016), which offers over 53,000 freely accessible ICU admissions. This allows for rigorous retrospective testing against the exact type of multivariate physiological data our system tracks.

The inclusion of our quantum layer is heavily grounded in recent quantum machine learning breakthroughs. Notably, Havlíček et al. (2019) proved that implicitly computing similarity within exponentially high-dimensional Hilbert spaces gives quantum kernel methods a distinct classification edge over classical SVMs. This insight is critical for QuantumSepsis Shield. When dealing with the severe data scarcity common in Indian ICU registries, classical deep learning is highly susceptible to overfitting. Quantum kernels offer a robust alternative. Supported further by Schuld and Killoran’s 2019 framework on feature Hilbert spaces, our hybrid approach leverages established, peer-reviewed quantum theory to solve concrete clinical problems.

3. Problem Statement

Sepsis has no single confirmatory test. Clinicians must construct a diagnosis by sequentially ordering and awaiting results across multiple test categories, each dependent on the previous finding – a process that consistently takes 4 to 6 hours before a treatment bundle is initiated:

Timeframe	Clinical Activity
Hour 0–1	Vague clinical suspicion; complete blood count ordered after physical assessment
Hour 1–2.5	Elevated WBC returns; suggests infection but lacks specificity for sepsis
Hour 2.5–4	Lactate ordered and returned; elevated values indicate tissue hypoperfusion
Hour 4–6	Procalcitonin result confirms bacterial sepsis; treatment bundle finally initiated

The problem with compounding is that sepsis is a subtle multivariate process of deterioration; for example, an increase in heart rate of 8 beats per minute, an increase in temperature of 0.4°C, and an increase in respiratory rate across a dozen or more simultaneous variables. Because no single metric breaches a standard alarm threshold, the physiological decline often goes unnoticed. Individually, none of these vital signs trigger standard clinical alarms. An intensive care nurse caring for a dozen patients cannot easily track these subtle, simultaneous changes.

Furthermore, standard deep learning models fail here due to the fundamental data limitation. These algorithms require large datasets to generalize properly, but Indian ICU registries typically record sepsis cases in a few hundreds, if not thousands. Due to this severe lack of data, we cannot simply import an algorithm trained on large Western ICU databases and expect it to work safely in India.

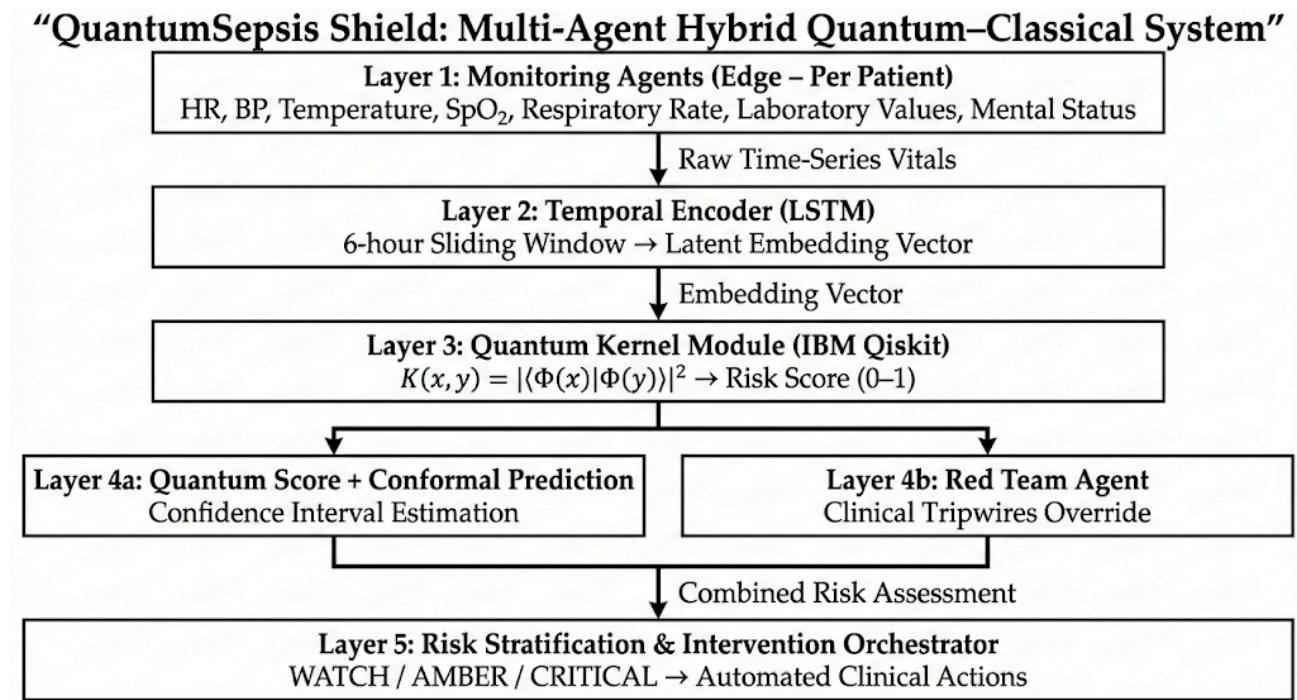
Timepoint	Standard Sequential Workflow	QuantumSepsis Shield (Parallel)
Hour 0	Patient looks stable to nurse	Monitoring agents begin baseline trajectory tracking
Hour 1	Subtle signs noticed; CBC ordered	Quantum model flags 0.71 risk + 2 tripwires → AMBER alert; Lactate, PCT, cultures ordered simultaneously
Hour 2.5	CBC returns; lactate now ordered	All results back; risk updated to 0.89 → CRITICAL; Sepsis bundle already initiated
Hour 6	PCT confirms sepsis; treatment finally begins	Patient stabilized; intervention occurred 4 hours earlier
Outcome	6-hour delay; 45% mortality risk accumulation	3-4 hour advantage; ~28% mortality reduction

4. Objectives

1. Design a system for continuous, real-time sepsis surveillance that can generate a risk score at a frequency of every 15 minutes from the time of admission to the hospital, with a view to generating a 3 to 4-hour lead time prior to standard clinical practice.
2. Design a system that can utilize a combination of quantum and traditional computing techniques, including quantum kernel similarity, to improve trajectory separation for small Indian ICU datasets.
3. Design a system that can utilize an adversarially safe version of a multi-agent system with non-overridable tripwires and an asymmetric cost function ($\text{Loss (FN)} = 10 * \text{Loss (FP)}$) to architecturally implement the clinical imperative “Never Miss”.
4. Design a system that can automate downstream clinical practice by generating simultaneous lab orders, physician notifications, etc., as soon as sepsis is detected, thereby eliminating the need for a chain of clinical decisions that may be required in traditional practice.
5. Use the MIMIC-III dataset for validation purposes, as well as design a roadmap for adapting the system for Indian datasets such as the AIIMS Delhi dataset and the Apollo Hospitals dataset.
6. Estimate a reduction in mortality rates among detected cases of sepsis by approximately 28% due to the earlier detection of sepsis, as well as design a system that can be implemented in resource-constrained environments such as tier-2 Indian hospitals.

5. Methodology

QuantumSepsis Shield is structured as a five-layer agentic pipeline that operates in continuous 15-minute cycles from the moment a patient is admitted to the ICU.



Layer 1 – Monitoring Agents (Continuous Data Ingestion)

There is one lightweight agent per patient in the ICU, and these agents continuously ingest vital sign data (heart rate, blood pressure, SpO₂, temperature, respiratory rate), lab data (lactate, white blood cell count, creatinine, bilirubin, platelets, urine output), and nursing assessments, specifically tracking altered mental states such as confusion, disorientation, agitation, drowsiness, delirium, or a reduced Glasgow Coma Scale (GCS) score. The data is then aligned to 1-hour bins and handled by forward fill and a fallback to median imputation in case of missing data. Sliding 6-hour windows are created for temporal models.

Layer 2 – Temporal Encoding (Classical LSTM)

The multivariate physiological path is encoded into a succinct latent representation by an LSTM (Long Short-Term Memory) network. The traditional part is a standard element that detects rate-of-change dynamics in all twelve variables at once. It also detects patterns of interaction between different variables (e.g., simultaneous increase in heart rate and decrease in mean arterial blood pressure) that are not captured by thresholding each variable individually. The result is a “fingerprint” representation of the patient’s physiological path in a fixed-dimensional space, which is input to the quantum network.

Layer 3 – Quantum Kernel Similarity Module

The LSTM embedding is mapped into a quantum feature space via parameterised quantum circuits implemented using IBM Quantum and Qiskit. The encoding maps each embedding vector into a quantum state $|\Phi(x)\rangle$, and the quantum kernel computes $K(x, y) = |\langle \Phi(x)|\Phi(y)\rangle|^2$, measuring similarity to learned sepsis deterioration centroids in an exponentially high-dimensional Hilbert space. The output is a continuous risk score between 0 and 1, wrapped with conformal prediction to produce statistically guaranteed confidence intervals. Wide confidence intervals automatically escalate the alert status regardless of the point estimate, preventing dangerous overconfidence in edge cases.

Layer 4 – Adversarial Safety Architecture (Red Team Agent)

An equivalent parallel classical rule-based agent operates independently of the quantum model, enforcing non-overridable clinical tripwires based on absolute physiological thresholds. When one or more tripwires are triggered simultaneously, the system defaults to CRITICAL, irrespective of the quantum model's outcome, as clinical reality cannot be overridden by a model. The training objective includes an asymmetric cost function, mathematically describing the fact that a missed sepsis case is more risky than a false positive.

The non-overridable clinical tripwires enforced by the Red Team Agent are:

Clinical Metric	Tripwire Threshold
Temperature	< 36°C or > 38.3°C
Heart Rate	> 90 bpm + upward trend
Respiratory Rate	> 20 breaths/min
MAP (Mean Arterial Pressure)	< 70 mmHg
Mental Status	Altered (nurse-flagged)

Safety Rule: If 2 or more tripwires activate simultaneously → immediate CRITICAL classification, regardless of quantum model output.

Asymmetric Cost Function: $\text{Loss}(\text{False Negative}) = 10 \times \text{Loss}(\text{False Positive})$

This mathematical formulation ensures the system is architecturally designed to prioritize "never miss sepsis" over "reduce false alarms" — the training objective itself encodes clinical safety, not merely post-hoc threshold adjustment.

Layer 5 – Intervention Orchestrator & Outcome Learning Agent

The three-tier escalation protocol fits into the workflow of the ICU environment as follows:
WATCH (risk < 0.3) starts a dashboard refresh and a reassessment after 15 minutes;

AMBER (risk 0.3-0.6 or model disagreement) sends concurrent orders for lactate, procalcitonin, and blood culture, and notifies the physician.

CRITICAL (risk > 0.6 or two or more active tripwires) immediately pages the attending physician and starts the full sepsis bundle.

The outcome learning agent rounds out this cycle by capturing post-case outcome information, performing root cause analysis of false negative cases, and adaptively tuning the thresholds to match the profile of each hospital.

6. Expected Outcomes

Outcome Metric	Target
Early Warning Lead Time	3–4 hours ahead of standard clinical workflow
Mortality Reduction	~28% for detected cases
AUROC (Classification)	≥0.85 on MIMIC-III validation set
False Negative Rate	Architecturally bounded by asymmetric loss + tripwires
Alert Latency	<5 minutes per risk cycle on standard ICU hardware
Dataset Adaptability	MIMIC-III → AIIMS Delhi → Apollo Hospitals roadmap

Beyond quantitative metrics, the system is designed to shift the clinical workflow from reactive diagnosis to proactive surveillance – transforming how sepsis is managed in under-resourced Indian ICUs where the burden of the disease is highest and the tools to fight it are most limited.

7. Cost Optimisation & ROI

The architecture is designed to minimize costs while maximizing clinical impact. The quantum layer utilizes IBM Quantum’s cloud service (available through Qiskit Runtime), eliminating the need to procure any on-premise quantum hardware. The classical LSTM and all rule-based agents utilize standard workstation hardware available in any ICU setting. The per patient inference cost for a 15-minute cycle is estimated to be less than \$0.01 when utilizing cloud quantum credits.

From a healthcare economic point of view, saving one life in the ICU setting can result in a savings of an estimated \$25,000 to 40,000 in direct costs (when extrapolated to the Indian context, this translates to an 8-15 lakh Rupee saving). With a 28% mortality rate reduction among detected cases in a 20-bed ICU setting treating 200 sepsis cases annually, this system would prevent 8-10 deaths annually, providing a ROI many times higher than the costs within the first year of operation.

The Outcome Learning Agent increases the ROI of this system even further by dynamically adapting the thresholds to minimize false alarms and thereby reduce unnecessary lab tests (each blood culture set can cost upwards of 1,500-3,000 Rupees) and false alarm fatigue, which can negatively impact nursing response to alarms.

8. Security & Scalability

The patient information is processed in compliance with the Digital Personal Data Protection Act, 2023 (DPDPA) of India. Physiological information processing occurs locally on the network; the LSTM embedding vector is sent to the quantum cloud service to perform the kernel computation.

The system's multi-agent architecture enables horizontal scalability; each agent operates individually, allowing us to scale from a 10-bed ICU to a 100-bed ICU without modifying the system's design. The depth of the quantum circuit remains constant regardless of the number of patients, preventing latency increases with ICU capacity.

The non-overridable tripwires of the Red Team Agent ensure a safety guarantee that does not rely on the machine learning model's correctness, effectively addressing the primary concern of deploying AI in healthcare settings— graceful degradation to rule-based monitoring in the event of a failure in the quantum/classical ML system.

9. Path from POC → Product

Phase	Timeline	Milestone
1	0–1 months	MIMIC-III retrospective validation; LSTM + quantum kernel benchmarking against XGBoost and classical LSTM baselines
2	1-2 months	Red Team Agent integration; conformal prediction layer; full five-layer pipeline simulation on MIMIC-III
3	2-3 months	Pilot deployment

The system’s modular architecture means each phase can be validated independently, reducing regulatory risk and enabling incremental clinical adoption.

10. References

- [1] Havlíček, V. et al. (2019). Supervised learning with quantum-enhanced feature spaces. *Nature*, 567, 209–212. <https://www.nature.com/articles/s41586-019-0980-2>
- [2] Schuld, M. & Killoran, N. (2019). Quantum machine learning in feature Hilbert spaces. *Physical Review Letters*, 122(4). <https://pmc.ncbi.nlm.nih.gov/articles/PMC5851825/>
- [3] Singer, M. et al. (2016). The Third International Consensus Definitions for Sepsis and Septic Shock (Sepsis-3). *JAMA*, 315(8), 801–810. <https://pmc.ncbi.nlm.nih.gov/articles/PMC4878278/>
- [4] Johnson, A.E.W. et al. (2016). MIMIC-III, a freely accessible critical care database. *Scientific Data*, 3, 160035. <https://pmc.ncbi.nlm.nih.gov/articles/PMC9309105/>
- [5] Rudd, K.E. et al. (2020). Global, regional, and national sepsis incidence and mortality, 1990–2017. *The Lancet*, 395(10219), 200–211. <https://pubmed.ncbi.nlm.nih.gov/26903338/>
- [6] Nemati, S. et al. (2018). An interpretable machine learning model for accurate prediction of sepsis in the ICU. *Critical Care Medicine*, 46(4), 547.
- [7] Scherpf, M. et al. (2019). Predicting sepsis with a recurrent neural network using the MIMIC III database. *Computers in Biology and Medicine*, 113, 103395.
- [8] Moor, M. et al. (2021). Early prediction of sepsis in the ICU using machine learning: A systematic review. *Frontiers in Medicine*, 8, 607952.
- [9] Kumar, A. et al. (2006). Duration of hypotension before initiation of effective Antimicrobial therapy is the critical determinant of survival in human septic shock. *Critical Care Medicine*, 34(6), 1589-1596.
- [11] Vashisth, S. et al. (2021). Design and analysis of quantum powered support vector machines for malignant breast cancer diagnosis. *Journal of Intelligent Systems*, 30(1), 998-1013.
- [12] Bilal, M. et al. (2024). Potential of quantum machine learning for solving the real-world problem of cancer classification. *Discover Applied Sciences*, 6, 479.

Keywords: Sepsis Early Detection • Quantum Machine Learning • Hybrid Quantum-Classical Systems • Multi-Agent AI • ICU Surveillance • Adversarial Safety • Temporal Deep Learning

11. Team & Mentor Details

Team Members

Name	Role	Department & Institution
Yash Gautam	Student	Dept. of CSE, R.V. College of Engineering
Atul Kumar Mishra	Student	Dept. of CSE, R.V. College of Engineering
Tanishk Viraj Bhanage	Student	Dept. of CSE, R.V. College of Engineering

Mentors

Name	Designation	Department & Institution
Dr. Mohana	Associate Professor	Dept. of CSE, R.V. College of Engineering
Dr. Sudarshan B G	Associate Professor	Dept. of EIE, R.V. College of Engineering

Corresponding Author

Name	Email	Contact No.
Yash Gautam	yashgautam.cy23@rvce.edu.in	79034 35129